SSV Discovery

Main Project Report

Submitted to the Mahatma Gandhi University, Kottayam

in partial fulfillment of requirements for the award of degree

Bachelor of Science

in

Computer Science

by

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POST GRADUATE DEPARTMENT OF COMPUTER SCIENCE

SREE SANKARA VIDYAPEETOM COLLEGE

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2024 - 25

CERTIFICATE

This is to certify that the report entitled **SSV Discovery** submitted by **Abhilash Roy** (220021025938) to Department of Computer Science in partial fulfillment of the B.Sc. degree in **Computer Science** is a bonafide record of the Main Project work carried out by him under my guidance and supervision. This report in any form has not been submitted to any other University or Institute for any purpose.

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Internal Examiner

External Examiner

Date Of Examination ....

DECLARATION

I hereby declare that the main project report **SSV Discovery**, submitted for partial fulfillment of the requirements for the award of degree of Bachelor of Science in Computer Science of the Mahatma Gandhi University, Kottayam, Kerala is a bonafide work done by me under supervision of Dr.Manusankar C.

This submission represents my own ideas in my own words and where ideas or words of others have been included, I have adequately and accurately cited and referenced the original sources.

I also declare that I have adhered to ethics of academic honesty and integrity and have not misrepresented or fabricated any data or idea or fact or source in my submission.I understand that any violation of the above will be a cause for disciplinary action by the institute and/or the University and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been obtained. This report has not been previously formed the basis for the award of any degree, diploma or similar title of any other University.

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Abstract

SSV Discovery is a web-based recruitment platform developed for SSV College to streamline campus hiring. It connects recruiters with students through a centralized system that features structured profiles, skill-based filtering, and natural language search powered by spaCy NLP. The platform simplifies the recruitment process by eliminating manual resume screening and providing direct access to verified student talent.

Built using Django and MySQL, it offers a user-friendly interface for both students and recruiters. SSV Discovery enhances placement opportunities while bridging the gap between academia and industry. It also provides a secure and scalable system that ensures data integrity and ease of maintenance. The platform aims to empower students by showcasing their skills and projects effectively, making them more visible to potential employers.

Acknowledgement

I take this opportunity to express my deepest sense of gratitude and sincere thanks to everyone who helped me to complete this work successfully. I express my sincere thanks to **Prof.(Dr.) Sudhakaran K.M.**, our Principal and management, for providing me with all the necessary facilities and support.

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I would like to express our sincere gratitude to **Asha Anand**, our class tutor and **Sri. Narayanan Namboothiri S** our technical assistant and all the teachers for their support, guidance and co-operation.

Finally I thank my family, and friends who contributed to the successful fulfilment of this project work.

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Contents

List of Figures

Chapter 1

Introduction

In today's competitive job market, campus recruitment plays a crucial role in connecting talented students with potential employers. However, traditional hiring methods such as manual resume screening, job fairs, and general-purpose job portals often fall short when it comes to efficiently discovering fresh graduate talent. These methods are typically time-consuming, inconsistent, and lack the necessary filtering to identify specific skill sets.

To address these challenges, we have developed SSV Discovery — a customized web-based recruitment platform exclusively for SSV College. This platform is designed to streamline the hiring process by offering structured student profiles, natural language search powered by NLP (spaCy), and precise filtering options based on skills, projects, and academic experience.

The system bridges the gap between students and recruiters by providing a centralized and verified database of student profiles, eliminating unnecessary complexity and delays. SSV Discovery not only simplifies the recruitment workflow but also empowers students to present their qualifications and capabilities in a professional and impactful way.

Chapter 2

Literature Review

Several job portals like LinkedIn, Indeed, and Naukri provide job listings and candidate searches. However, these platforms lack the ability to perform natural language searches effectively at a college level. Traditional recruitment platforms rely heavily on keyword-based searches, whereas SSV Discovery leverages NLP techniques to provide more accurate and relevant candidate recommendations. The integration of AI-driven search with SQL-based filtering makes SSV Discovery a unique solution for the academic recruitment domain.

Furthermore, most existing platforms are not tailored to the specific needs of fresh graduates, often overlooking student projects, academic achievements, and college-specific credentials. SSV Discovery addresses this gap by focusing exclusively on verified student profiles from SSV College, ensuring authenticity and relevance. By combining structured academic data with intelligent search features, the platform significantly enhances the discoverability of student talent for recruiters looking to hire freshers.

Chapter 3

Key Features and Functionality of SSV Discovery

SSV Discovery has been designed to cater to the specific needs of recruiters, students, and administrators within an academic environment. Each user type benefits from a tailored set of features that aim to streamline the recruitment process, enhance user engagement, and maintain the integrity and functionality of the platform. The features are categorized below for clarity:

3.1 Recruiter Features

- Natural Language Search Capability: Recruiters can perform searches using everyday language rather than being restricted to keyword-based queries. For example, a recruiter can type ”Computer Science students with Python and machine learning experience” and get accurate, filtered results.

- Advanced Filters for Precision: After submitting a query, recruiters can narrow down the results using filters based on criteria such as skills, years of experience, academic qualifications, and preferred locations. This ensures that recruiters find the most relevant candidates quickly.

- Anonymous Profile Viewing: To promote transparency while respecting privacy, recruiters can view student profiles anonymously without needing to log in. This encourages recruiters to explore potential candidates freely without friction.

- No Mandatory Login: The platform eliminates unnecessary barriers by allowing recruiters to browse and evaluate profiles without account creation, making the system more accessible and recruiter-friendly.

3.2 Student Features

- Profile Registration and Management: Students can sign up and build a detailed profile that includes personal information, educational background, skills, certifications, and achievements.

- Showcase Projects and Experience: The platform enables students to highlight their academic and personal projects, internships, and work experience. This offers recruiters a more comprehensive view of each candidate's capabilities.

- Resume and Profile Picture Upload: Students can upload their resumes and profile images, giving recruiters visual and document-based insights into the candidate's qualifications and professionalism.

- Enhanced Visibility in Searches: The use of NLP ensures that students' profiles appear in recruiter searches based on actual relevance to the query, increasing their chances of being shortlisted.

3.3 Admin Features

- Platform Monitoring and Analytics: Administrators have access to dashboards that allow them to monitor user activity, search trends, and general platform usage. This data can help improve system performance and content moderation.

- User Account Management: Admins can manage student and recruiter accounts, verify identities, and delete or flag inappropriate content to maintain a safe and professional environment.

- Search Trend Analysis: Admins can view frequently searched terms and recruiter preferences. This insight can help colleges tailor their curriculum or training programs to match market demand.

- System Maintenance and Settings: The admin panel also includes options to adjust backend settings, manage database entries, and ensure overall platform security and integrity.

Overall, the SSV Discovery platform offers a rich and functional set of features that empower recruiters to find suitable candidates with ease, enable students to present themselves effectively, and give administrators the tools needed to manage and maintain the platform efficiently. Its modular design ensures that the system remains scalable and adaptable for future enhancements and wider institutional adoption.

Chapter 4

System Configuration

4.1 Software Configuration

1. **Operating System:**
   - Windows 10/11
   - Ubuntu 20.04+
   - macOS (for development)
2. **Backend:**
   - Python 3.7 or higher
   - Django 5.0.2
   - MySQL Server 8.0+
   - spaCy (for NLP)
   - English Language Model: `en-core-web-sm` or `en-core-web-md`
3. **Frontend:**
   - HTML5
   - CSS3
   - JavaScript
   - Bootstrap 5 (for responsive UI)

4. **Additional Tools:**
   - Visual Studio Code (or any IDE like PyCharm)
   - Git (for version control)
   - pip (Python package manager)
   - Virtualenv (for Python environments)
   - MySQL Workbench or phpMyAdmin (optional GUI for database)
5. **Python Libraries:**
   - `spacy`
   - `mysqlclient` or `pymysql` (for DB connection)
   - `django-crispy-forms` (optional for better form UI)

4.2 Hardware Configuration

1. **For Development Machine:**
   - Processor: Intel i5 (8th Gen) / AMD Ryzen 5 or higher
   - RAM: Minimum 8 GB (Recommended: 16 GB for smooth NLP model handling)
   - Storage: At least 256 GB SSD (for faster boot and build times)
   - Display: 13" or larger (Full HD recommended)
2. **For Hosting/Deployment (if self-hosted):**
   - Server: Ubuntu 20.04 LTS (cloud-based or local server)
   - CPU: 2-core or higher
   - RAM: 4 GB minimum (8 GB preferred)
   - Disk Space: 50 GB minimum
   - Internet: Stable broadband connection for serving web pages

Chapter 5

Development Tools and Technologies used

5.1 Django 5.0.2 (Python)

Django is a high-level web framework built using Python, designed to promote rapid development and clean, pragmatic design. The version used in this project, Django 5.0.2, includes the latest features, security patches, and performance optimizations. Django simplifies the backend development process by offering a built-in ORM for database operations, an admin panel for content management, built-in user authentication, middleware support, URL routing, and a robust templating engine. These features help in rapidly building secure and scalable web applications. The Django framework acts as the backbone of this project, managing user requests, handling data processing, and rendering dynamic web pages to users. Its modular design also makes it easier to maintain and extend the application with minimal effort.

5.2 HTML5, CSS3, JavaScript (Bootstrap 5)

The frontend of the application is crafted using a combination of HTML5, CSS3, and JavaScript, along with the Bootstrap 5 framework. HTML5 is used to define the structure and semantics of the web pages, supporting multimedia and modern browser capabilities. CSS3 adds styling to the interface, providing layout control, animations, and responsive design through media queries. JavaScript introduces interactivity, allowing users to experience real-time validations, dynamic form handling, and interactive UI elements. Bootstrap 5 further enhances the user interface by offering pre-designed UI components like buttons, modals, navbars, and forms, which are mobile-first and responsive by default. This stack ensures a smooth, clean, and intuitive user experience for both students and recruiters.

5.3 Database: MySQL

MySQL is a reliable, open-source relational database management system (RDBMS) used to store structured data. It is known for its speed, security, and support for complex queries, making it an ideal choice for web applications. In this project, MySQL is used to manage tables such as users, student profiles, recruiter information, search logs, and administrative settings. It integrates seamlessly with Django through its ORM (Object Relational Mapping) system, allowing developers to interact with the database using Python code instead of raw SQL. MySQL ensures data integrity, efficient indexing for faster searches, and support for transactions, which are crucial for storing and retrieving candidate information reliably.

5.4 Natural Language Processing (NLP):spaCy

spaCy is a modern, fast NLP library in Python used for processing and analyzing text data. It supports tasks such as tokenization, part-of-speech tagging, dependency parsing, named entity recognition, and word vector similarity. In this project, spaCy's en-core-web-sm model is used to interpret recruiter input queries in natural language. For example, if a recruiter types "Python developer with 2 years of experience," the system can extract relevant keywords, skills, and experience levels using spaCy and match them with appropriate student profiles. This enables smarter and more context-aware searching compared to traditional keyword-based systems, greatly improving the accuracy and relevance of candidate suggestions.

5.5 Hosting Deployment: Self-hosted Linux Server

The project is deployed on a self-hosted Linux server, which provides a secure, stable, and cost-effective platform for hosting web applications. Linux servers are widely used in production environments due to their performance, reliability, and open-source nature. Deploying on a self-hosted server allows full administrative control over software packages, firewall settings, and server behavior. This method is suitable for academic projects, as it eliminates the need for commercial hosting services. Additionally, Linux offers compatibility with Python and Django, simplifying deployment and maintenance processes. It also supports secure remote access via SSH, enabling easy updates and monitoring.

5.6 Version Control: Git and GitHub

Version control is essential in managing collaborative software development. Git, a distributed version control system, helps track changes to source code, maintain different versions, and collaborate with team members efficiently. GitHub is a cloud-based platform for hosting Git repositories and offers features like issue tracking, pull requests, and branch management. In this project, Git is used to maintain clean and structured code, revert back to previous versions if needed, and ensure that changes are documented. GitHub facilitates collaboration among team members by providing a centralized codebase, enabling peer code reviews and continuous integration. This setup significantly improves code quality, productivity, and teamwork.
Repository Url:
`https://github.com/AbHilaz/ssv_discovery.git`

Chapter 6

System Analysis

6.1 Existing System

The traditional hiring process involves:

- Manual searching through resumes.
- Limited search functionalities in job portals.
- Keyword-based filtering rather than context-based matching.
- Lack of personalized ranking systems.

These limitations result in inefficient candidate discovery and increased hiring time.

6.2 Proposed System

SSV Discovery enhances the recruitment process by:

- Providing natural language search capabilities.
- Allowing recruiters to filter candidates based on skills, experience, and location.
- Offering an intuitive web-based interface for students to create profiles.
- Enabling anonymous recruiter browsing.
- Implementing an admin dashboard for platform monitoring and management.

6.3 Feasibility Analysis

Feasibility analysis is the process of evaluating the viability of a project based on technical, economic, operational, and other factors. For the SSV Discovery platform, the feasibility is analyzed under the following categories. This step is crucial to ensure that the project can be successfully implemented within available resources and time constraints. It also helps in identifying potential risks, estimating costs, and determining the overall practicality and sustainability of the solution. By examining these aspects early in the development cycle, stakeholders can make informed decisions and align the project goals with real-world capabilities.

6.3.1 Technical feasibility

The technical feasibility of the SSV Discovery platform evaluates whether the current technology, tools, and infrastructure are sufficient to support the development and deployment of the application. The platform is built using a reliable and modern technology stack, including Django (Python-based web framework) for the backend, MySQL for database management, and frontend technologies like HTML5, CSS3, Bootstrap, and JavaScript. These technologies are well-supported, widely used, and compatible with most systems, ensuring long-term maintainability and scalability. Additionally, the integration of natural language processing using spaCy enhances the search capabilities, making the platform more intuitive for recruiters. The development team has the required technical expertise to implement and maintain these components, and the system can be hosted on standard web servers or cloud platforms like AWS or Heroku. Therefore, the project is considered technically feasible within the available resources and timeframe.

6.3.2 Economic Feasibility

The economic feasibility of the SSV Discovery project is highly favorable due to its low development and operational costs. There are no licensing fees involved, as the project is built using open-source tools and frameworks such as Django, MySQL, and spaCy. The development can be carried out using personal systems, significantly minimizing the initial investment. For deployment, the platform can be hosted on free-tier cloud services, making it cost-effective for demonstration and testing purposes. As usage grows, the system can be scaled based on available budget and institutional needs. The maintenance cost is also minimal, with only optional expenses for domain registration and server hosting, making the project economically sustainable in the long run.

6.3.3 Operational Feasibility

The operational feasibility of the SSV Discovery platform is strong, as it is designed to be highly user-friendly and easy to navigate. The use of Bootstrap ensures a clean and intuitive user interface, allowing both students and recruiters to interact with the system seamlessly without the need for technical expertise. Minimal training is required for end users, with only a brief onboarding session necessary for administrators to manage the platform effectively. Additionally, the platform is tailored to fulfill a specific functional need—bridging the gap between college students and recruiters through intelligent, NLP-driven search features—making it a well-suited and practical solution for academic recruitment.

6.3.4 Legal and Ethical Feasibility

The legal and ethical feasibility of the SSV Discovery platform is sound, as it adheres to responsible data handling practices by storing only verified student information and avoiding the collection of excessive or sensitive personal data. The system prioritizes user privacy and maintains transparency in data usage. Additionally, all tools and frameworks employed in the development of the project are open-source and comply with permissive licenses, making them ideal for educational use without violating any intellectual property rights. Overall, there are no significant legal or ethical concerns that would hinder the implementation or deployment of this platform.

6.3.5 Schedule Feasibility

The schedule feasibility of the SSV Discovery platform is well-aligned with typical academic timelines. The project is structured to be completed within a standard academic semester, accommodating key phases such as requirement gathering, design, development, testing, and deployment. Given the scope and modular design of the platform, a small team of 2–4 developers with intermediate proficiency in Django and Python can effectively manage and execute the project within the allotted timeframe. With proper planning and task distribution, the project milestones can be met without undue pressure, making it feasible from a scheduling perspective.

6.4 Data Flow Diagram

A Data Flow Diagram (DFD) or a bubble chart is a graphical tool for structured analysis. It was De Macro in 1978 and Gene and Carson in 1979 who introduced DFD. DFD models a system transforms the data and creates, output data-flows which go by suing external entities from which data flows to a process which to other processes or external entities or files. Data in files many also flow to processes as inputs. There are various symbols used in DFD.

Bubbles represent the process. Named arrows indicate the dataflow. External entities are represented by rectangles and are outside the system such as vendors or customers with whom the system interacts. They either supply or consume data. Entities supplying data are known as sources and those that consume data are called sinks. Data are stored in a data store by a process in the system. Each component in a DFD is labelled in with a descriptive name. Process names are further identified with a number. DFD can be hierarchically organized, which help In partitioning and analyzing large systems. As a first step, one Data Flow Diagram Can depict an entire system, which gives the system overview. It is called Context.

Diagram of level 0 DFD. The context diagram can be further expanded. The successive expansion of DFD from the context diagram that giving more details is known as levelling of DFD. Thus of top down approach is used, starting with an overview and then working out the details. The main merit of DFD is that it can provide an overview of what data a system would process, what transformation of data are done, what filesare used, and where the result flow. The data flow diagram of Post Office Management System has been represented as a hierarchical DFD context level DFD was drawn first. Then the processes were decomposed into several elementary levels and are represented in the order of importance.

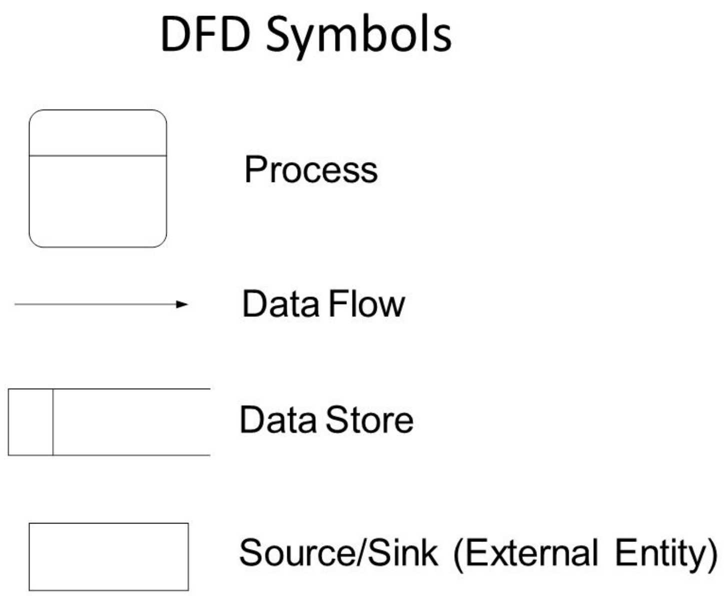

Figure 6.1: DFD Symbols

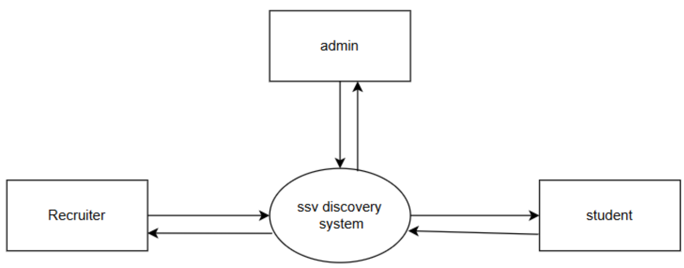

Figure 6.2: Level 0

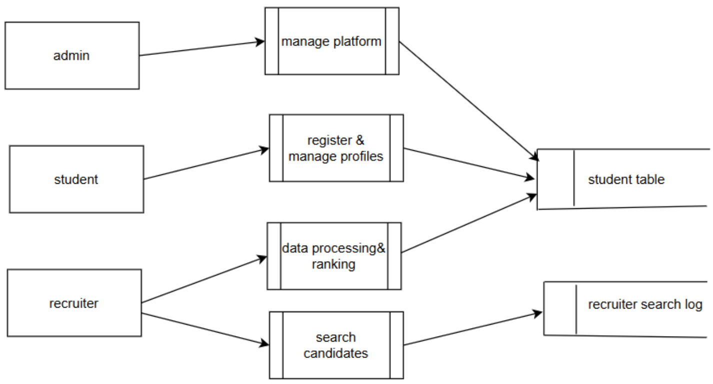

Figure 6.3: Level 1

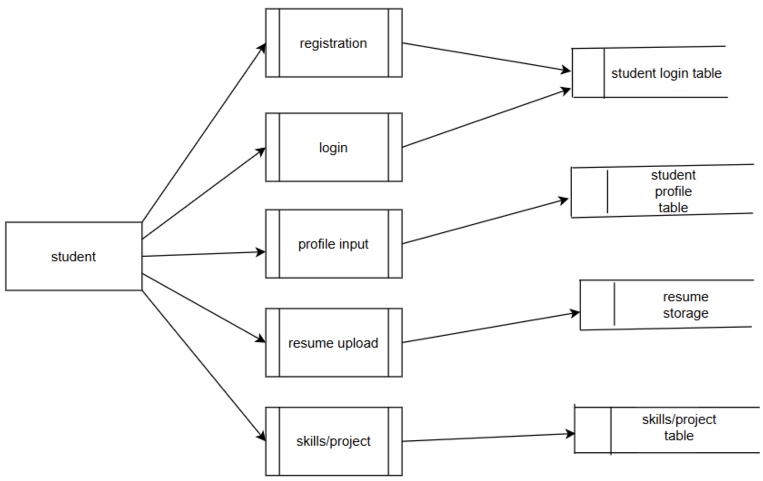

Figure 6.4: Level 2

Chapter 7

System Design

7.1 Input Design

The input design of the SSV Discovery platform focuses on ensuring accurate, user-friendly, and secure data entry from various user types, including students, recruiters, and administrators. Students provide inputs such as personal details, skills, academic background, project descriptions, and resumes through structured forms. These forms include validation mechanisms to ensure data consistency and completeness. Recruiters input search queries using natural language or filters like skill, location, and experience level, which are processed through the system's NLP engine. Administrators input platform settings and user management commands via a secure admin panel. The input design emphasizes clarity, simplicity, and responsiveness to enhance user interaction and reduce errors during data entry.

7.2 Output Design

The output design of the SSV Discovery platform is centered around presenting meaningful, relevant, and easy-to-understand results to its users. For recruiters, the primary output is a list of student profiles that match their search queries—these results are enhanced using Natural Language Processing (NLP) techniques to ensure semantic relevance rather than just keyword matching. Each result includes key student information such as name, skills, academic background, and resume download links, all displayed in a clean, card-style layout. For students, the output includes updates on recruiter visits, interest in their profile, and application statuses. Administrators receive dashboards summarizing system analytics, such as user activity, search trends, and platform performance metrics. The overall output design ensures that each user type receives actionable and well-organized information, contributing to a smoother and more efficient recruitment process.

7.3 Database Design

The database design of the SSV Discovery platform is structured to efficiently support both student and recruiter interactions through multiple interconnected tables. The Users table stores essential details of all users, including students and recruiters, serving as a central point of authentication and identification. The Profiles table holds comprehensive student information such as resumes, skill sets, academic history, and work experience, enabling effective profile presentation to recruiters. The Recruiters table contains data related to the companies and their representatives who use the system to find suitable candidates. To support data-driven insights, the Search Logs table anonymously records recruiter search behavior, which can be utilized for analytics and platform improvement. Finally, the Admin table is responsible for managing system settings and controlling access privileges, ensuring secure and smooth operation of the platform. This structured database design ensures data consistency, scalability, and seamless integration across various user types and functionalities.

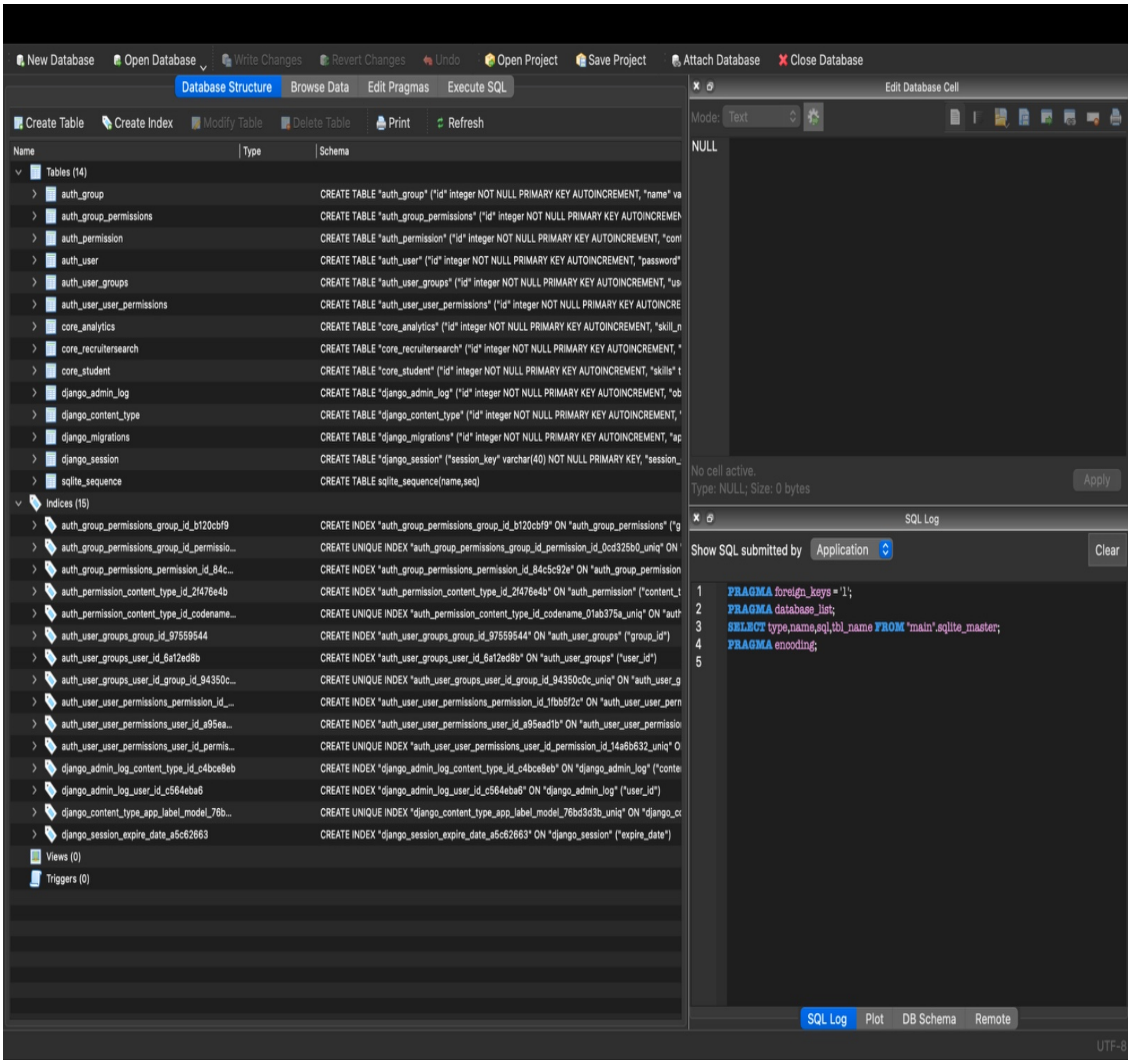

Figure 7.1: Database

7.3.1 Table Design

The table design of the SSV Discovery project is structured to support a smooth and efficient talent discovery process for students and recruiters. The database includes essential tables such as Users, Profiles, Recruiters, Search Logs, and Admin. Each table is normalized to reduce redundancy and improve data integrity. Fields are appropriately typed , and constraints like primary keys (PK), foreign keys (FK), and unique keys (UNI) are used to ensure consistency. This structured schema allows for secure authentication, accurate profile matching, and efficient querying, making the overall system both scalable and maintainable.

Field	Type	Null	Key	Default	Extra
user_id	INT	NO	PRIMARY	NULL	AUTO_INCREMENT
name	VARCHAR(100)	NO		NULL	
email	VARCHAR(100)	NO	UNIQUE	NULL	
password_hash	TEXT	NO		NULL	
role	ENUM('student', 'recruiter', 'admin')	NO		student	
created_at	TIMESTAMP	NO		CURRENT_TIMESTAMP	
last_login	DATETIME	YES		NULL	

Figure 7.2: User Table

Field	Type	Null	Key	Default	Extra
profile_id	INT	NO	PRIMARY	NULL	AUTO_INCREMENT
user_id	INT	NO	FOREIGN	NULL	
degree	VARCHAR(100)	NO		NULL	
branch	VARCHAR(100)	NO		NULL	
year_of_passing	YEAR	NO		NULL	
experience_years	FLOAT	YES		0	
resume_file_path	VARCHAR(255)	YES		NULL	
profile_picture_path	VARCHAR(255)	YES		NULL	
bio	TEXT	YES		NULL	

Figure 7.3: Student Profile

Field	Type	Null	Key	Default	Extra
skill_id	INT	NO	PRIMARY	NULL	AUTO_INCREMENT
profile_id	INT	NO	FOREIGN	NULL	
skill_name	VARCHAR(100)	NO		NULL	

Figure 7.4: Skills Table

Field	Type	Null	Key	Default	Extra
project_id	INT	NO	PRIMARY	NULL	AUTO_INCREMENT
profile_id	INT	NO	FOREIGN	NULL	
project_title	VARCHAR(150)	NO		NULL	
description	TEXT	YES		NULL	
tools_used	VARCHAR(200)	YES		NULL	
duration	VARCHAR(50)	YES		NULL	

Figure 7.5: Project Table

Field	Type	Null	Key	Default	Extra
recruiter_id	INT	NO	PRIMARY	NULL	AUTO_INCREMENT
user_id	INT	NO	FOREIGN	NULL	
organization_name	VARCHAR(100)	NO		NULL	
designation	VARCHAR(100)	YES		NULL	
contact_number	VARCHAR(15)	YES		NULL	

Figure 7.6: Recruiters Table

Field	Type	Null	Key	Default	Extra
search_id	INT	NO	PRIMARY	NULL	AUTO_INCREMENT
recruiter_id	INT	NO	FOREIGN	NULL	
search_query	TEXT	NO		NULL	
timestamp	TIMESTAMP	NO		CURRENT_TIMESTAMP	
filters_applied	TEXT	YES		NULL	

Figure 7.7: Search Logs

Field	Type	Null	Key	Default	Extra
admin_id	INT	NO	PRIMARY	NULL	AUTO_INCREMENT
user_id	INT	NO	FOREIGN	NULL	
permissions	TEXT	YES		NULL	

Figure 7.8: Admins Table

Field	Type	Null	Key	Default	Extra
feedback_id	INT	NO	PRIMARY	NULL	AUTO_INCREMENT
user_id	INT	NO	FOREIGN	NULL	
message	TEXT	NO		NULL	
submitted_at	TIMESTAMP	NO		CURRENT_TIMESTAMP	

Figure 7.9: Feedback Table

Chapter 8

Testing and Implementation

8.1 System Testing

The testing and validation phase of the SSV Discovery platform was a crucial part of the development lifecycle, ensuring that the application functions reliably, performs efficiently, and provides a secure and user-friendly experience for both students and recruiters. A wide range of testing techniques were implemented to examine various aspects of the platform, covering both functional and non-functional requirements.

8.1.1 Functional Testing

This type of testing was carried out to validate whether the core features of the platform were operating as expected. Key modules such as user registration, login/logout processes, profile creation and editing, recruiter search functionality, and admin panel operations were all tested thoroughly. The testing included valid input scenarios to ensure correct outputs and invalid input cases to check for proper error handling. Edge cases, such as empty form submissions, incorrect credentials, and duplicate entries, were tested to assess the robustness of form validations and the application's ability to handle unexpected behavior without crashing.

8.1.2 Performance Testing

The efficiency of the application, especially the search functionality powered by Natural Language Processing (NLP), was tested under varying loads. Search queries were analyzed for their average response time and how well the platform performed under multiple simultaneous users. This helped identify potential bottlenecks in the system, particularly in the ranking algorithm and database query execution. Stress testing was also performed using mock data to determine the system's response when processing a large number of student profiles, ensuring scalability for future expansion.

8.1.3 Security Testing

Ensuring the safety of user data was a top priority. The Django backend provides several built-in security features, which were verified during this phase. Tests were conducted to check for common vulnerabilities such as SQL injection, cross-site scripting (XSS), cross-site request forgery (CSRF), and session hijacking. Authentication and authorization flows were tested to ensure users could not access or manipulate data that they were not permitted to view. Passwords were checked to be securely hashed and never stored in plain text. The system was also reviewed to ensure HTTPS compatibility and protection against data leakage.

8.1.4 User Experience (UX) Testing

A select group of users, including students and recruiters, were invited to use the platform and provide feedback on its usability. Testers evaluated how easy it was to navigate the UI, understand prompts, submit search queries, and manage profiles. Feedback was collected regarding visual design, feature placement, responsiveness, and accessibility. Based on this input, the frontend was fine-tuned to improve intuitiveness, reduce visual clutter, and ensure a seamless experience across different user roles.

8.1.5 Compatibility Testing

Given the wide variety of devices and browsers used today, compatibility testing ensured the system worked across different platforms. The application was tested on major browsers such as Google Chrome, Mozilla Firefox, Microsoft Edge, and Safari. Additionally, it was accessed on desktops, tablets, and mobile devices of different screen sizes to verify that the responsive design (built using Bootstrap 5) provided consistent functionality and appearance.

8.1.6 Regression Testing

With continuous improvements and feature updates, regression testing was performed after each code iteration to ensure that new features did not unintentionally break existing functionality. Automated tests were created for repetitive tasks, and manual testing was also conducted for dynamic components of the application.

8.2 Validation

The final stage involved validating the platform against the original project requirements and objectives. Each module and functionality was cross-checked to ensure it met the expected outcomes. The system was validated as a complete solution capable of helping recruiters find unique student profiles effectively using NLP-enhanced search capabilities. The rigorous testing and validation process helped build confidence in the stability, security, and usability of the SSV Discovery platform. With successful completion of multiple test cases and positive user feedback, the system is considered ready for deployment in a real-world academic setting. Continuous monitoring and updates can further enhance its performance and user satisfaction over time.

8.3 System Implementation

Implementation Details: The Django Backend Handles Authentication and Search Queries The core logic of the application is managed through Django, which is responsible for securely handling user authentication (login/logout) for both students and recruiters. Django's built-in authentication system allows for password hashing, session management, and user access control. In addition to authentication, Django handles recruiter search queries by interacting with the database and applying custom filtering logic. When a recruiter submits a search request, the backend processes the natural language input using NLP (spaCy), translates it into SQL-compatible filters, and retrieves matching student profiles from the database.

A Custom Ranking Algorithm Sorts Profiles Based on Recruiter Queries To improve the relevance of results, a custom ranking algorithm has been implemented. Instead of just listing all profiles that match a set of keywords, the algorithm analyzes the recruiter's query and evaluates student profiles based on various factors such as skill match, years of experience, and educational background. Profiles that closely align with the recruiter's intent are ranked higher in the results. This provides a smarter and more personalized recommendation system compared to simple keyword-based sorting.

The Frontend is Designed with a Minimal, User-Friendly UI: The frontend is crafted using HTML, CSS, JavaScript, and Bootstrap to ensure a clean and intuitive user experience. The layout follows minimal design principles to reduce clutter and make navigation easy. Recruiters can search profiles with a single input field, and students can view or update their profiles with clear prompts. The use of Bootstrap ensures the UI is responsive, adapting smoothly to different screen sizes including desktops, tablets, and mobile phones. The overall design prioritizes simplicity and accessibility.

8.3.1 System Maintenance

System maintenance is an essential phase in the software development life cycle that ensures the long-term stability, usability, and performance of the SSV Discovery platform after its deployment. As user requirements evolve and new challenges arise, maintaining the system becomes critical to keep it efficient, secure, and relevant. The maintenance activities for this project can be categorized into several types:

1. Corrective Maintenance

This type of maintenance involves fixing bugs or issues that may arise during real-time usage of the platform. Even after rigorous testing, certain edge-case scenarios or unforeseen errors might occur in the live environment. Corrective maintenance ensures that such issues are addressed quickly, such as fixing broken search filters, resolving login issues, or patching form validation errors. These fixes are applied promptly to maintain system reliability and user trust.

2. Adaptive Maintenance

Adaptive maintenance involves making changes to the system to accommodate new hardware, software environments, or policy changes. For example, if the institution decides to shift to a different cloud provider or update to a newer version of Python or Django, the system may require adaptation to remain compatible. This also includes modifying the system to support future upgrades, such as transitioning from MySQL to PostgreSQL or integrating with third-party APIs for external resume parsing.

3. Perfective Maintenance

This refers to enhancing the functionality and user experience of the platform based on user feedback and evolving requirements. Over time, recruiters and students may request additional features or improved performance. Perfective maintenance may involve improving the search algorithm, adding new filters for better profile discovery, refining the UI for better accessibility, or optimizing database queries for faster performance.

4. Preventive Maintenance

Preventive maintenance is carried out to reduce the risk of future system failures by proactively identifying and fixing potential issues before they cause problems. This includes tasks like code refactoring, database indexing, server log monitoring, regular backups, and upgrading dependencies to prevent security vulnerabilities. It also involves reviewing analytics to detect patterns that might signal upcoming problems.

5. Security Maintenance

Security is an ongoing concern, especially for platforms handling personal and academic data. Regular security audits should be performed to ensure that vulnerabilities are patched, access controls are up-to-date, and SSL certificates are renewed. Security maintenance also includes updating libraries and frameworks to their latest secure versions, enforcing stronger password policies, and monitoring for unauthorized access attempts.

6. Documentation Maintenance

Keeping technical documentation, user manuals, and help guides up to date is a part of ongoing system maintenance. As the system evolves, any changes to its features or workflows should be reflected in the documentation to support smooth onboarding of new users and developers.

Effective system maintenance ensures the SSV Discovery platform continues to function efficiently and meets the expectations of its users over time. By implementing a structured maintenance strategy that includes corrective, adaptive, perfective, preventive, security, and documentation maintenance, the system can remain robust, scalable, and secure in a real-world academic environment.

Chapter 9

Screenshots

The following screenshots illustrate key components and functionalities of the SSV Discovery platform. They provide a visual representation of how users interact with the system and help showcase the design, navigation flow, and usability of the application. The homepage features clean navigation with options for student and recruiter login, admin access, and user registration, all secured through password authentication. Once logged in, students can create and manage their profiles, while recruiters can use the search functionality powered by NLP to find relevant candidates.

Screenshots also include the search results page where recruiter queries return ranked student profiles based on skill match, experience, and academic background. The profile submission and update pages are intuitive and guide the student through uploading resumes, listing skills, and adding project experience. Additional visuals display the admin panel used to manage users and moderate submitted content.

These screenshots validate the working implementation of core features like authentication, profile management, recruiter search, and responsive UI design. Each screenshot supports the functional goals of the project and demonstrates that the platform is user-friendly, legally compliant, and built to assist academic institutions in smarter student-recruiter connectivity.

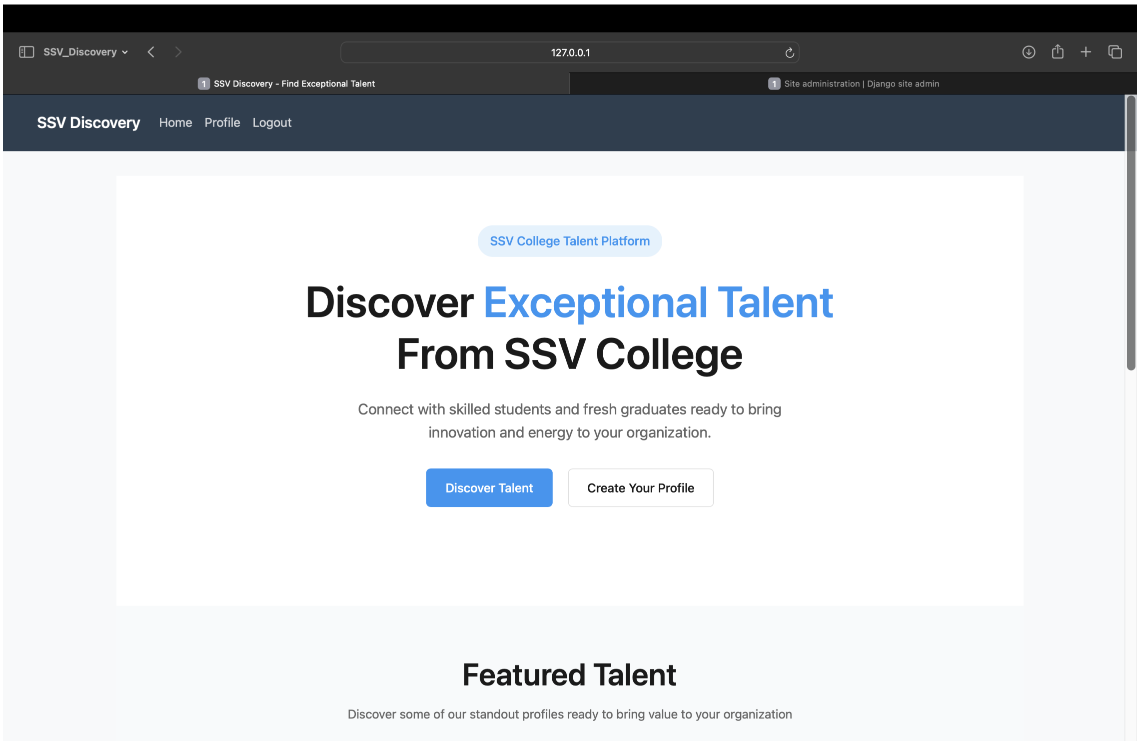

Figure 9.1: Home Page

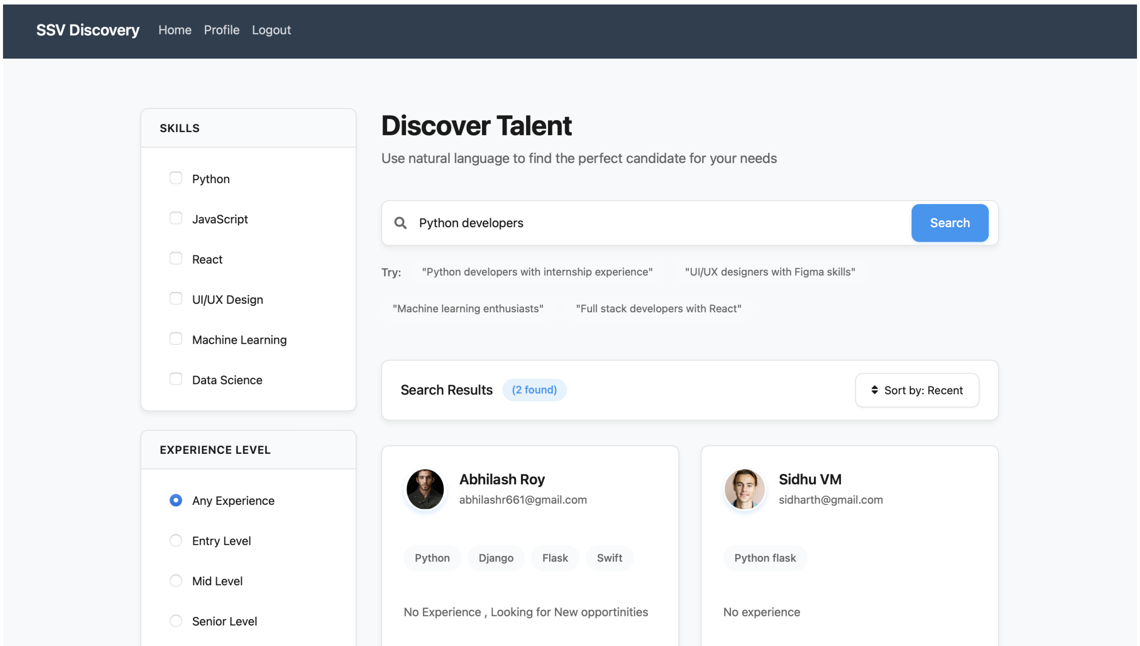

Figure 9.2: Search Page

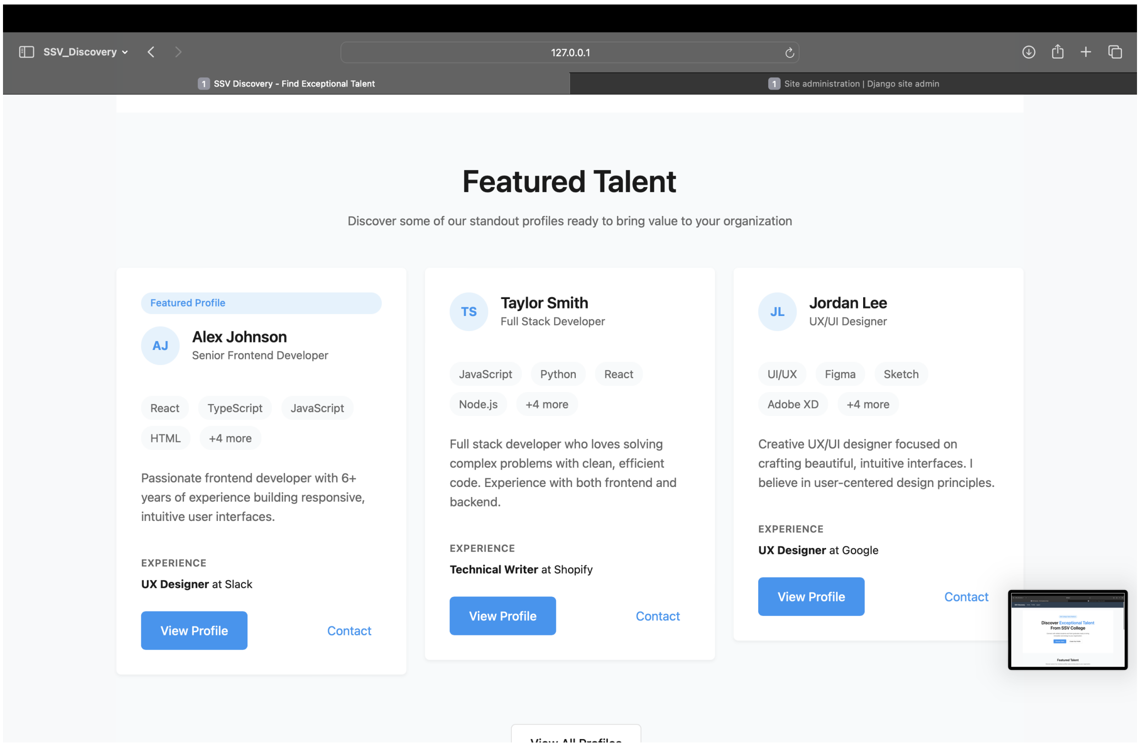

Figure 9.3: View Profile Page

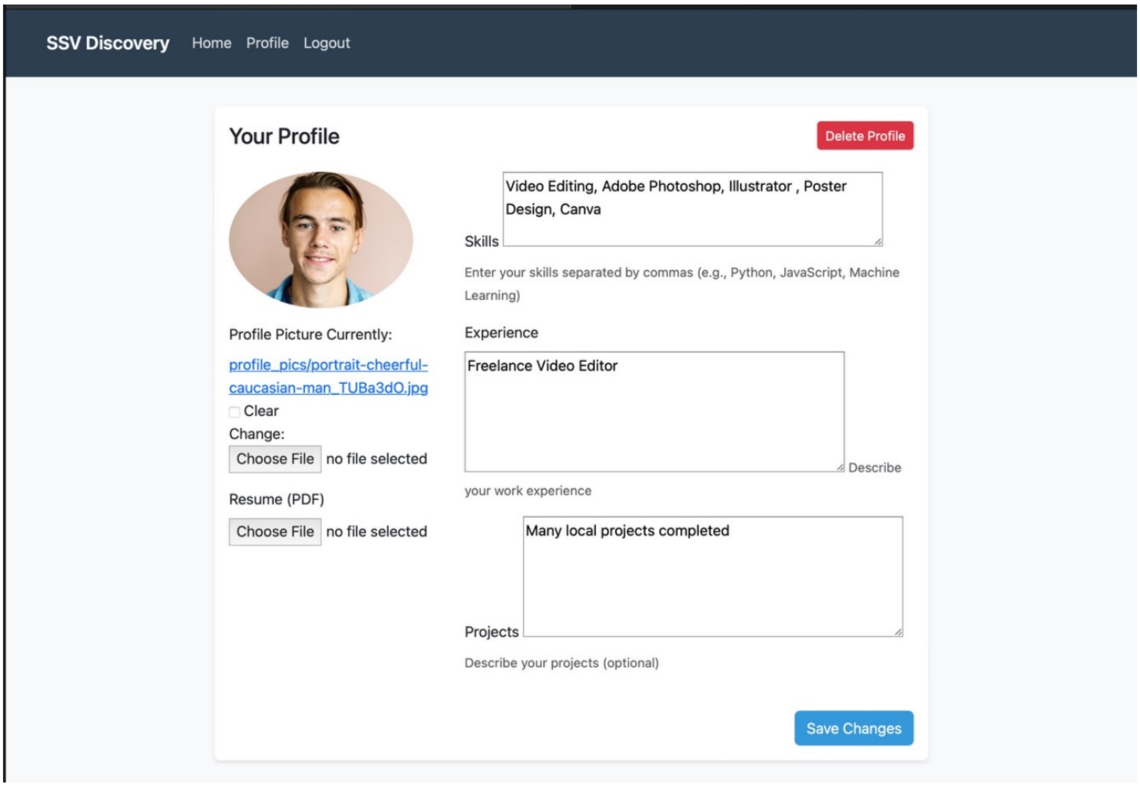

Figure 9.4: Your Profile Page

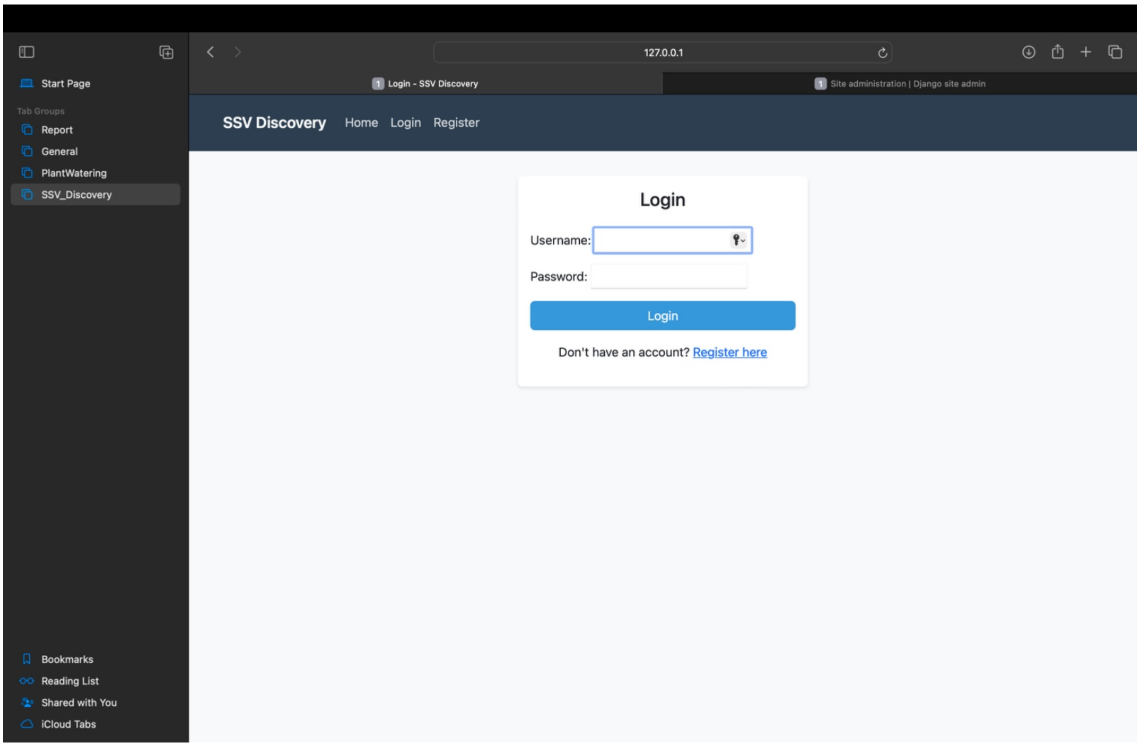

Figure 9.5: Login Page

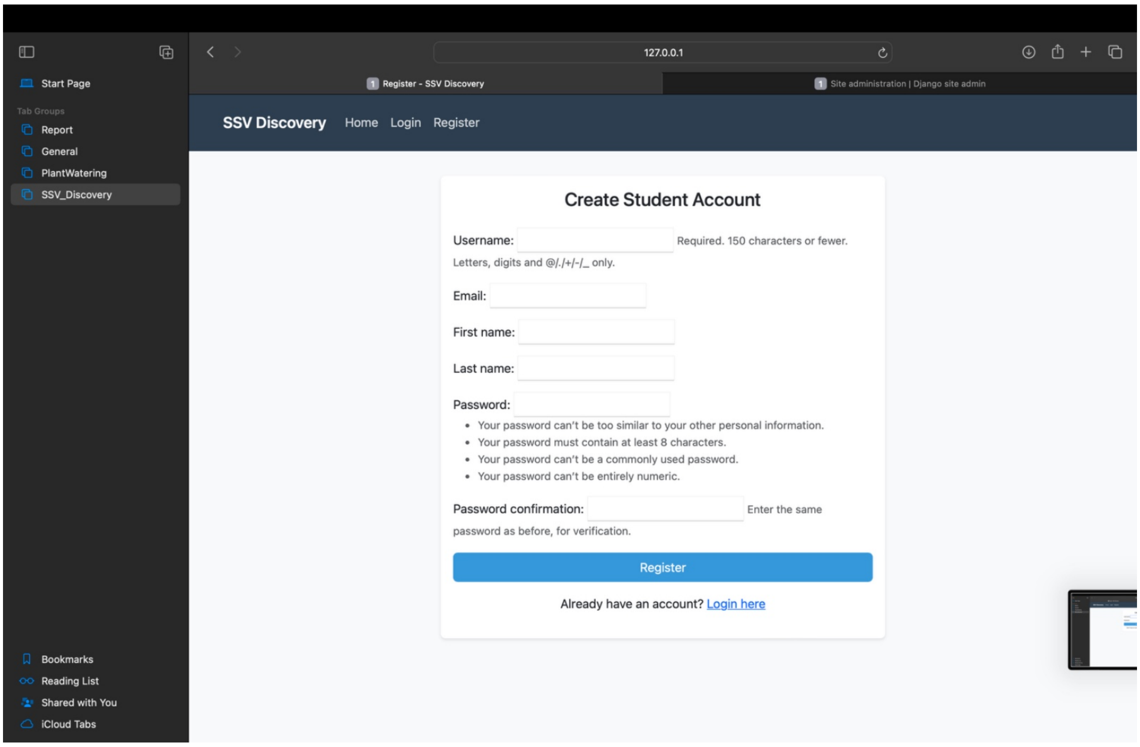

Figure 9.6: Student Account Creation Page

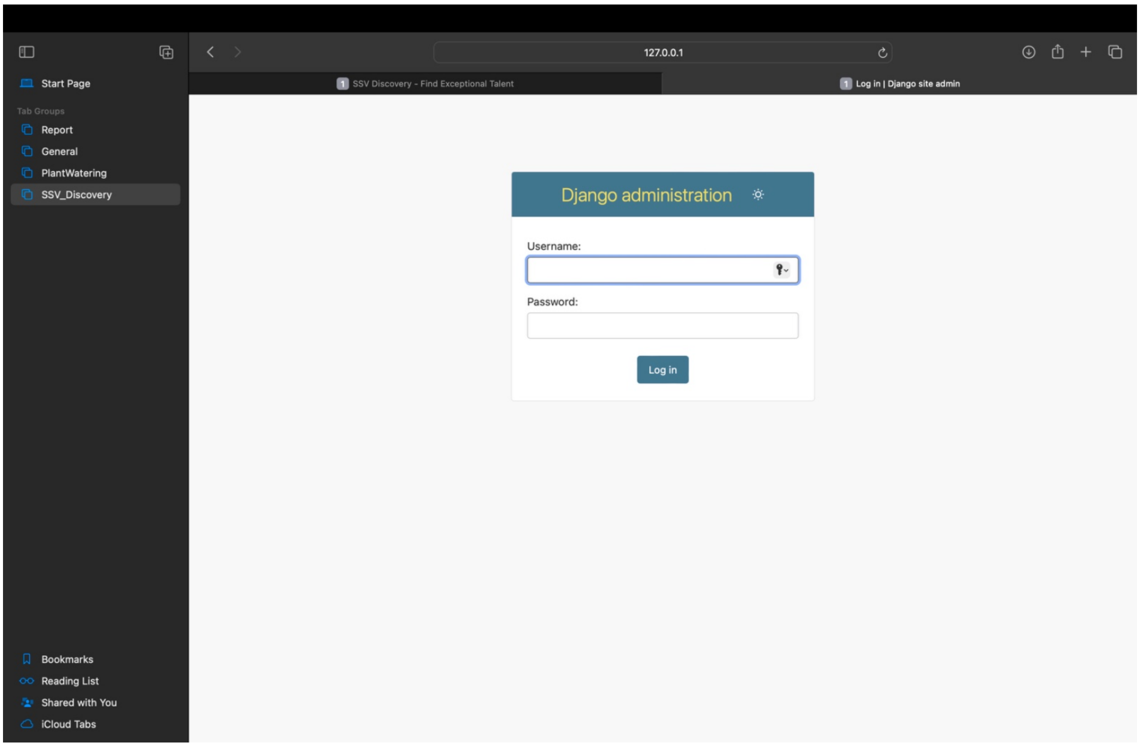

Figure 9.7: Admin Login Page

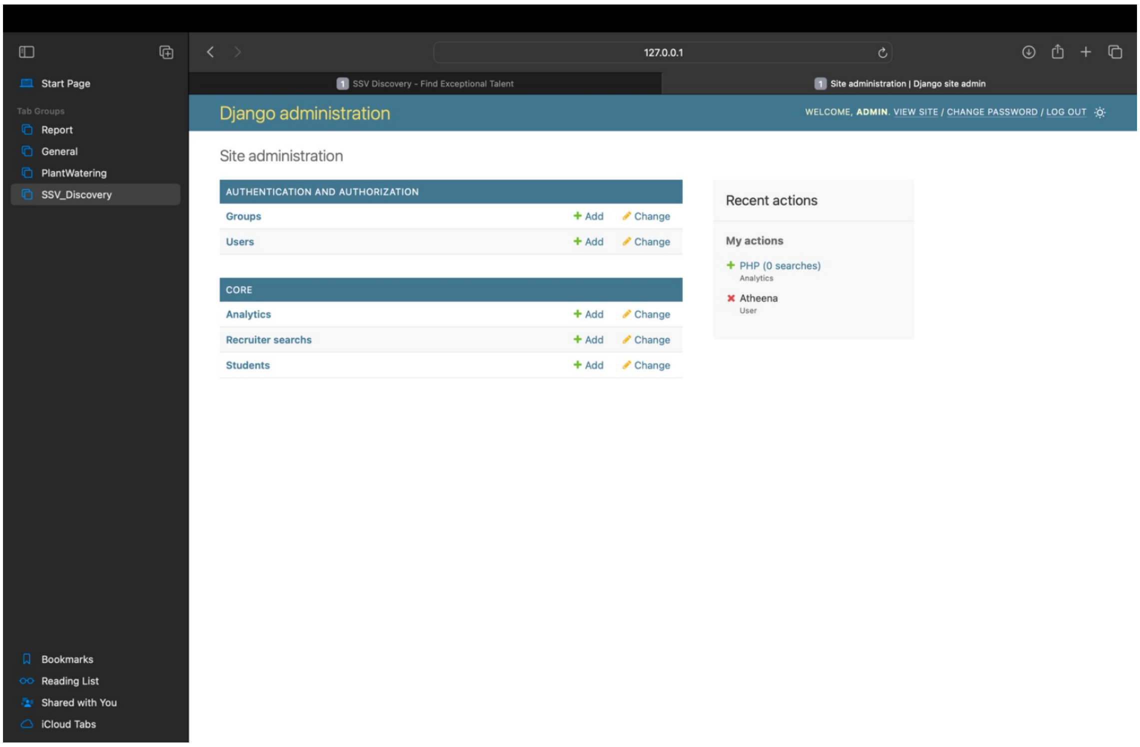

Figure 9.8: Admin Dashboard

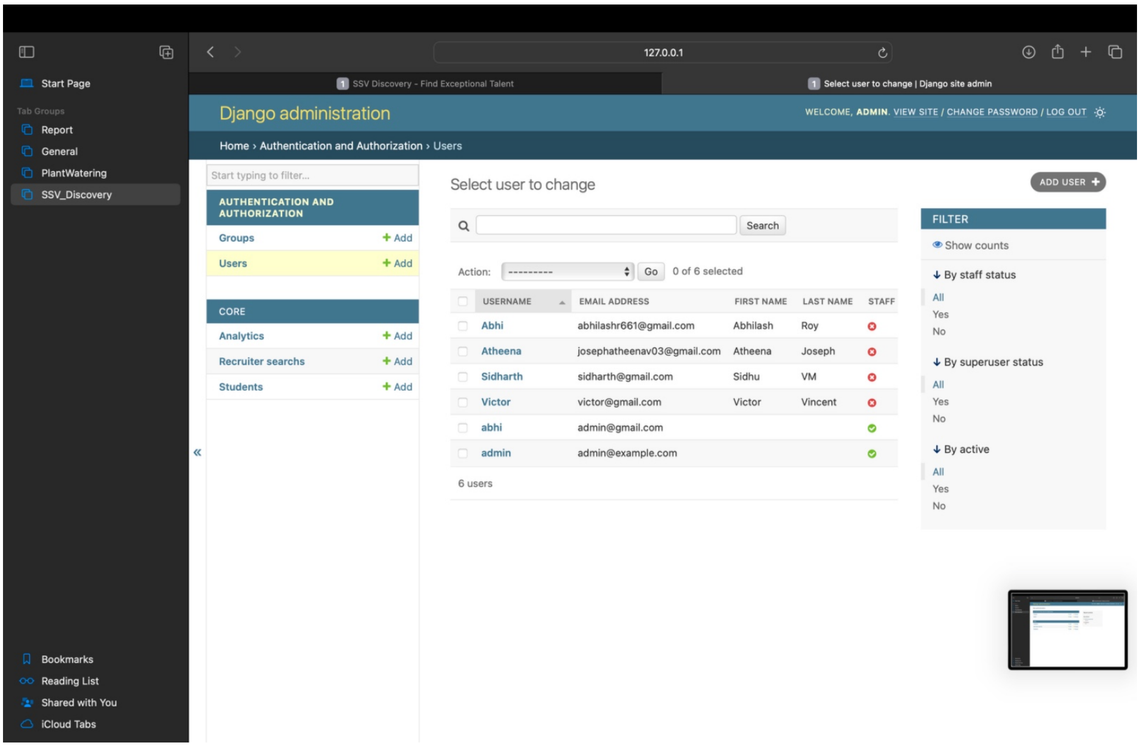

Figure 9.9: Admin's User Management Page

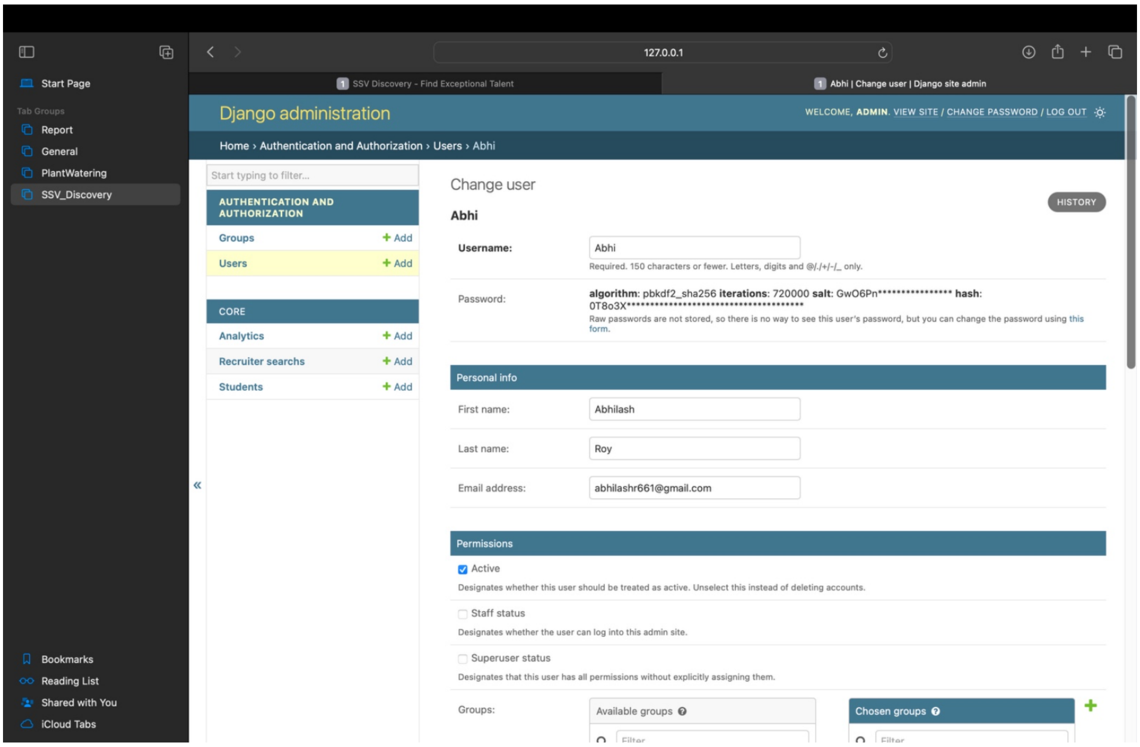

Figure 9.10: User Management Page

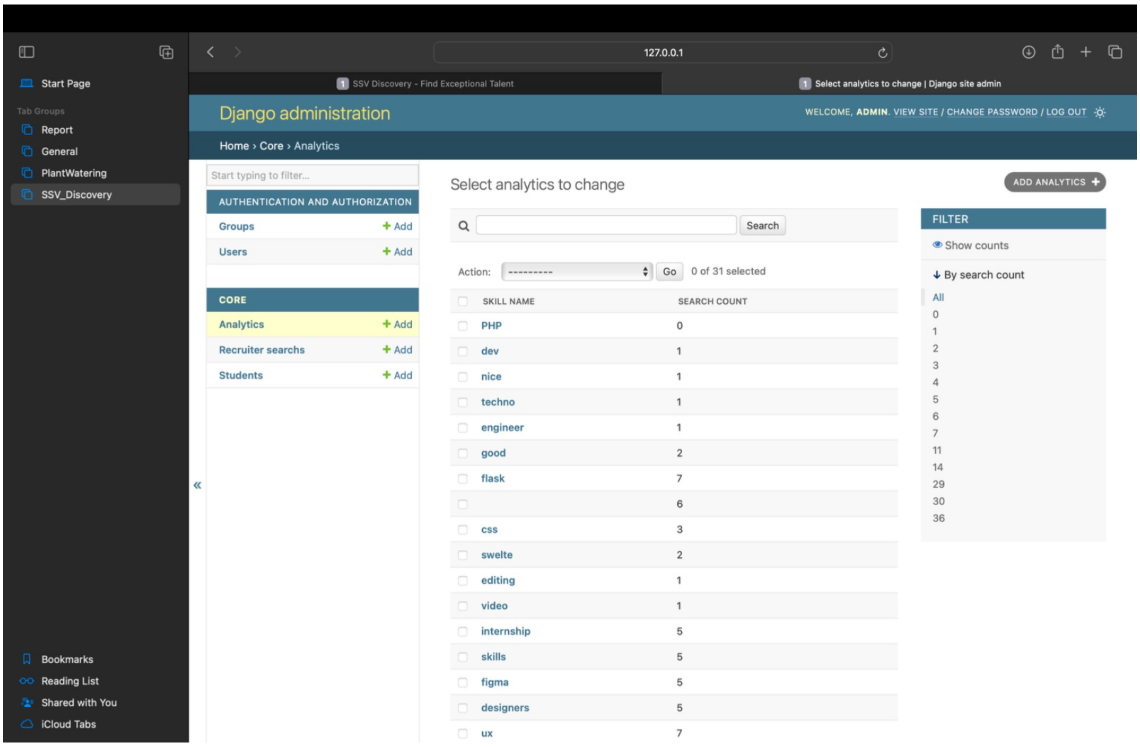

Figure 9.11: Recruiters Search Analytical Page

Chapter 10

Conclusion

SSV Discovery offers an innovative, seamless, and intelligent approach to campus recruitment by integrating artificial intelligence with natural language processing (NLP) capabilities. Designed specifically for SSV College, the platform bridges the gap between talented students and potential recruiters through a more personalized and efficient search experience. Unlike traditional keyword-based platforms, SSV Discovery utilizes NLP to understand recruiter intent and provides highly relevant profile matches, streamlining the selection process.

The user-friendly interface, built using modern web technologies such as Django, Bootstrap, and MySQL, ensures a smooth and responsive experience across various devices. Recruiters can perform smart searches, and students can maintain dynamic profiles — all within a secure and accessible environment. The system also includes admin-level controls to monitor activity and manage content, making it well-suited for academic institutions. In conclusion, SSV Discovery is not just a tool but a stepping stone towards smarter, AI-powered recruitment solutions tailored to academic ecosystems. It reflects a meaningful blend of technology and purpose, aiming to empower both students and recruiters with better opportunities.

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